

PROGRESS REPORT

1. Brief Introduction

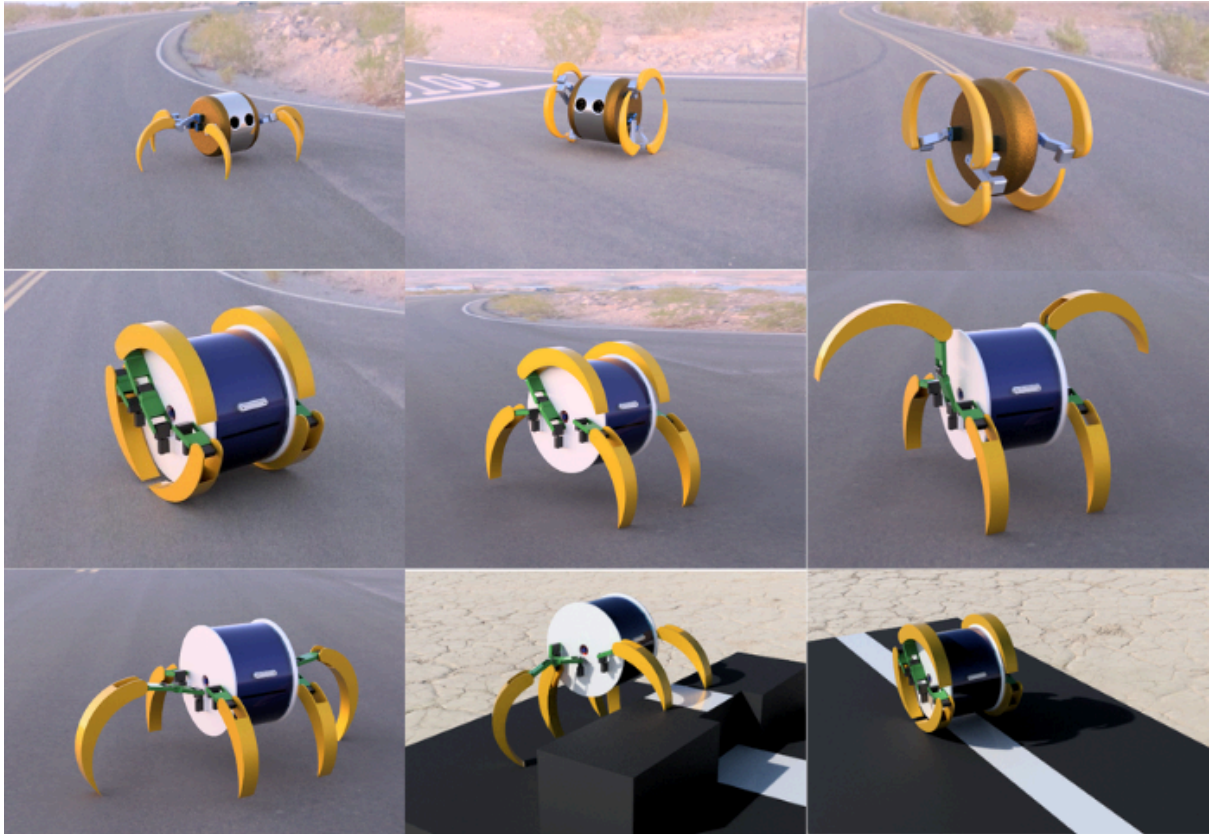


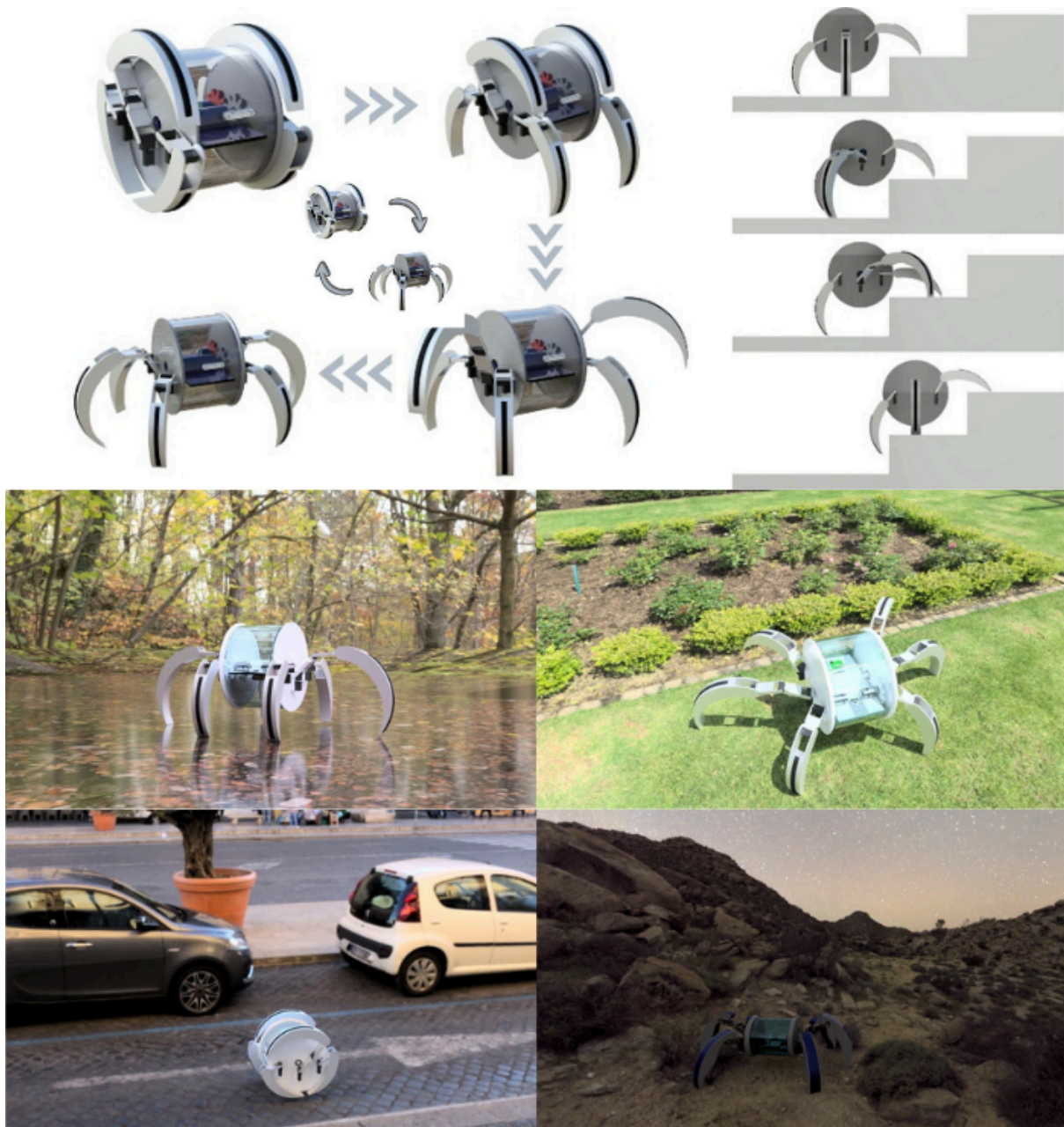
Fig1.1: Ideation Design

- **Rollopod** is a transforming hexapod robot.
- It features a unique dual-mode locomotion system, designed to switch between a six-legged walking mode and a high-speed rolling mode.
- **The walking mode** provides stability and adaptability, allowing the robot to navigate rough, uneven terrain and climb over obstacles.
- **The rolling mode** is designed to overcome the speed limitations of traditional hexapods, enabling efficient, fast movement on flat surfaces.
- A key innovation of this design is its ability to **continuously scan the environment using its sensors even while in rolling motion**, a feature market alternative lack. The robot is intended for multi-sector

applications requiring high adaptability, including search and rescue, industry exploration and planetary exploration.

2. Initial Concept & Idea Realization





3. Prototype 1.0: Initial 3D Printed Assembly



4. Prototype 2.0: Metal Chassis Design & Fabrication



5. Prototype 3.0 : Material and Design Improvement (Current)

